

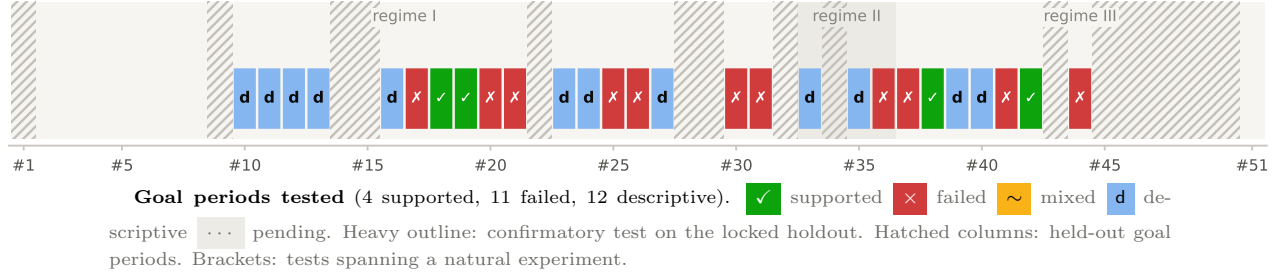
H31: Consensus time scales with the interaction graph's spectral gap

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

Complete 40% Faithfulness 1/5 Usefulness 2/5

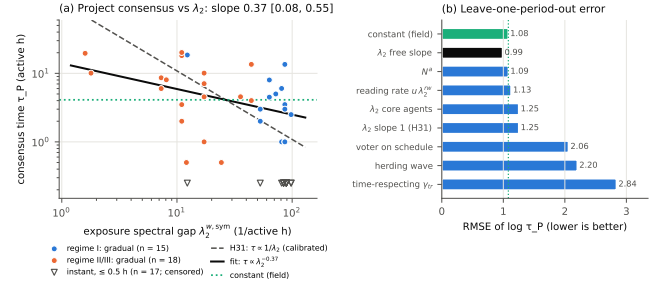
Bottom line. Consensus in the village does not follow the $1/\lambda_2$ law: gradual project consensus scales only as $\lambda_2^{-0.37}$ and is forecast no better than by a constant (≈ 4 active h, 80%: 1–18 h), while half of all consensus is set by the kickoff or arrives in one-window waves.

level: hypothesis · pipeline checks 86% · updated 2026-10-03



WHAT HAPPENED

Consensus events were defined by rule before outcomes and timed across non-holdout periods #10–#44: project-share majorities from H11’s labels (63 events, 20 periods), the #26 runoff vote, and content-alignment convergence. Their times were regressed on the exposure graph’s spectral gap λ_2 . Gradual project consensus (33 events, median 4.5 active h) scales sub-linearly, $\tau \propto \lambda_2^{-0.37}$ [0.08, 0.55], not as $1/\lambda_2$. The slope-1 law loses to a constant forecast out of sample (log-RMSE 1.25 vs 1.08); a free slope wins by 8%. The synthetic run on the real schedules had already put this test’s power at 0.13. Of the other events, 13 were frozen at the kickoff and 17 were one-window waves. Content never converged: it diverged after the kickoff in 12 of 35 blocks. The #26 runoff jumped in 0.76 h (post hoc). Holdout not run.



Observables. (a) Gradual project-consensus times against the exposure spectral gap, with the calibrated slope-1 law (dashed), the free fit (solid) and the constant forecast (dotted). Triangles: instant one-window events ($\tau \leq 0.5$ h, censored). (b) Leave-one-period-out error of each forecast rule. Only the free sub-linear slope beats the constant, by 8%. The voter, herding-wave and time-respecting models miss by an order of magnitude.

PHYSICAL MODEL

Model 10’s voter and Potts coarsening, plus graph diffusion (HH68). W_{ij} : agent j ’s messages first seen by i (call-start visibility), per active hour. L is its symmetrized Laplacian, with spectral gap λ_2 . The state x_i is an agent’s project (H11 labels) or its whitened statement embedding. Rivals run on the same reading schedule: DeGroot averaging (time-respecting gap γ_{tr}), the voter model ($\tau \propto N/u$, u = reading rate), a herding wave, and a field (kick-locked timing).

$$L = D - \frac{1}{2}(W + W^T), \quad 0 = \mu_1 < \lambda_2 \leq \mu_3 \leq \dots$$

$$\dot{x}_i = \sum_j W_{ij} (x_j - x_i) \Rightarrow \tau \propto 1/\lambda_2 \quad (\text{H31}), \quad \tau_{\text{voter}} \propto N/u$$

$$\log \tau_e = c + b \log(1/\lambda_2) + \epsilon_e : \quad \hat{b} = 0.37 [0.08, 0.55] \quad \text{vs.} \quad b = 1$$

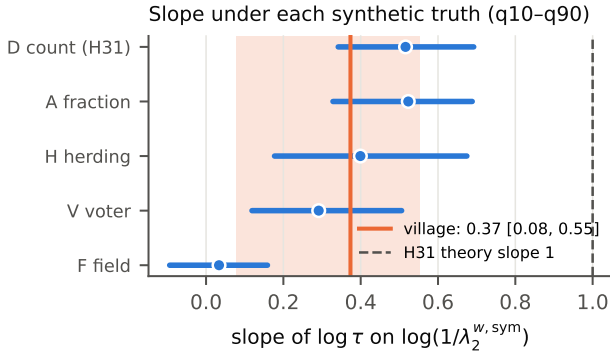
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly negative, with one usable number. A swarm operator can’t use the textbook “consensus time $\propto 1/\lambda_2$ ”. In chat rooms everyone sees everything, so λ_2 is mostly message volume. And consensus comes in three kinds: about a fifth is fixed by the kickoff prompt, a quarter arrives as one-window link-sharing waves, and the rest takes hours. The usable rule is a base rate: from a project’s first appearance, expect a majority after about 4 active hours (80%: 1–18 h). One lead is worth testing: rooms with poorly connected members converge more slowly (post hoc). Content alignment never converged; it decayed after the kickoff. So more talk did not buy agreement in content.

H31-consensus-time-spectral-gap · generated 2026-10-03 19:06 by infra/summaries/build_summaries.py · card hypotheses/H31-consensus-time-spectral-gap/README.md · ratings: H31 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 slope 1 on $\log(1/\lambda_2)$; constant	0.37 [0.08, 0.55]; RMSE 1.25 vs 1.08	failed
P2 no N slope	-0.51 [-1.38, 0.72]	holds (not identifiable)
P3 kick-locked	0.41 vs placebo 0.25 ($p = 0.002$), all from kickoff-frozen events	trivial
P4 $\geq 60\%$ abrupt rises	13% (45% with instant waves)	failed
P5 content slope on $1/\gamma_{tr}$	no convergence event (12 diverge)	untestable
P6 room sign as dif- fusion	3/4 two-room periods	weak pass
P8 synthetic power ≥ 0.5	0.13 for $\lambda_2^{w, sym}$	failed
P9 median τ 0.5–4 h	4.5 h	failed (narrowly)
P10 runoff ≤ 1 h	12.5 h by rule; 0.76 h rise post hoc	failed



Second figure. Synthetic truths run on the 28 real block schedules, through the same labels and detection rule. Every coupled dynamics gives an attenuated slope (0.3–0.5); a pure field gives about 0. The village estimate (orange band) excludes the field but cannot separate diffusion, voter and herding.

GOAL PERIODS IN DETAIL

G10 d descriptive
 G11 d descriptive
 G12 d descriptive
 G13 d descriptive
 G16 d descriptive
 G17 $\times$ failed
 G18 $\checkmark$ supported
 G19 $\checkmark$ supported
 G20 $\times$ failed
 G21 $\times$ failed
 G23 d descriptive
 G24 d descriptive
 G25 $\times$ failed
 G26 $\times$ failed
 G27 d descriptive
 G30 $\times$ failed
 +11 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Thirty-three gradual events from 14 periods, with one primary test that failed. The sub-linear slope ($p \approx 0.04$) survives regime, period-length and N controls. It does not survive the core-agent or event-window variants, dropping the pre-NE09 periods, or within-regime splits. λ_2 , message volume, N and regime co-vary. The split into frozen, instant and gradual events and the “uneven exposure” decomposition are post hoc. Labels measure attention (artifact mentions), not work. The call-start visibility rule is doubtful in regime I. The #26 onset rule fired at the period start. Content uses one embedding model at 30-min resolution.

ROUND 2

Holdout run (#29, #46, #47, #49, #50) after sign-off. An agent-level test: do poorly connected agents adopt later? Find what triggers instant waves (a shared link seen by ≥ 3 agents in one call cycle?). Use the #39→#40→#41 merge as an intervention.

No round-2 redirects yet.